Project Report 1

Cost-Efficient RAG (Retrieval-Augmented Generation) System

1. Project Title

Cost-Efficient Retrieval-Augmented Generation (RAG) System

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2. Project Overview

The Cost-Efficient RAG System is an AI-powered document question-answering application that retrieves relevant information from PDF documents before generating responses using a Large Language Model (LLM). The system minimizes hallucinations by grounding responses in retrieved document context while reducing inference costs through efficient semantic search.

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3. Objectives

- Build an intelligent document question-answering system.
- Retrieve only the most relevant document chunks.
- Reduce LLM token consumption.
- Generate citation-based responses.
- Create a scalable and modular AI application.

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4. Problem Statement

Traditional LLMs answer questions using only their training data, leading to outdated or hallucinated responses. The objective of this project is to improve factual accuracy by retrieving relevant document content before generating answers.

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5. Tech Stack

- Python
- FastAPI
- LangChain
- ChromaDB
- Sentence Transformers (all-MiniLM-L6-v2)
- Google Gemini 2.5 Flash
- PyPDF
- Docker

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6. System Architecture

```
      User
        |
        ▼
   FastAPI Backend
        |
   Upload PDF File
        |
   PDF Loader
        |
  Text Chunking
        |
Embedding Generation
        |
    ChromaDB
        |
    Retriever
```

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Prompt Builder

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Gemini 2.5 Flash

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Citation Formatter

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Final Response

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7. Workflow

1. Upload PDF.
2. Extract text.
3. Split text into chunks.
4. Generate embeddings.
5. Store embeddings in ChromaDB.
6. Retrieve relevant chunks.
7. Build prompt.
8. Generate answer using Gemini.
9. Return answer with citations.

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8. Key Features

- PDF ingestion
- Semantic search
- Citation support
- FastAPI REST APIs
- Vector database integration
- Modular architecture
- Cost-efficient inference

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9. API Endpoints

Endpoint	Description
/health	Check API status
/ingest	Upload and process PDF
/query	Ask questions

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10. Applications

- Enterprise Knowledge Base
- Research Assistant
- Educational Chatbot
- Internal Documentation Search
- Customer Support

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11. Future Enhancements

- Multi-document search
- Hybrid Search (BM25 + Vector Search)
- OCR Support
- Authentication
- Conversation Memory
- Multi-modal RAG

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12. Conclusion

The Cost-Efficient RAG System demonstrates modern Retrieval-Augmented Generation techniques using semantic search and vector databases to improve answer accuracy while minimizing LLM inference costs.